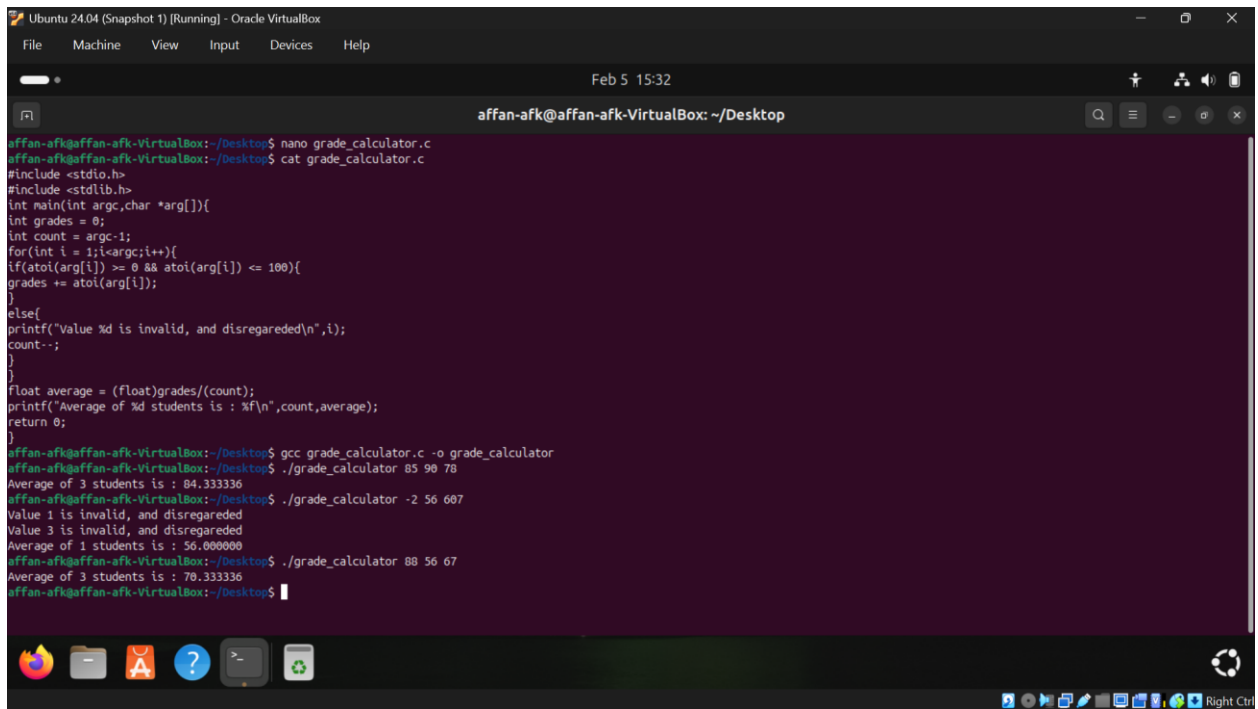
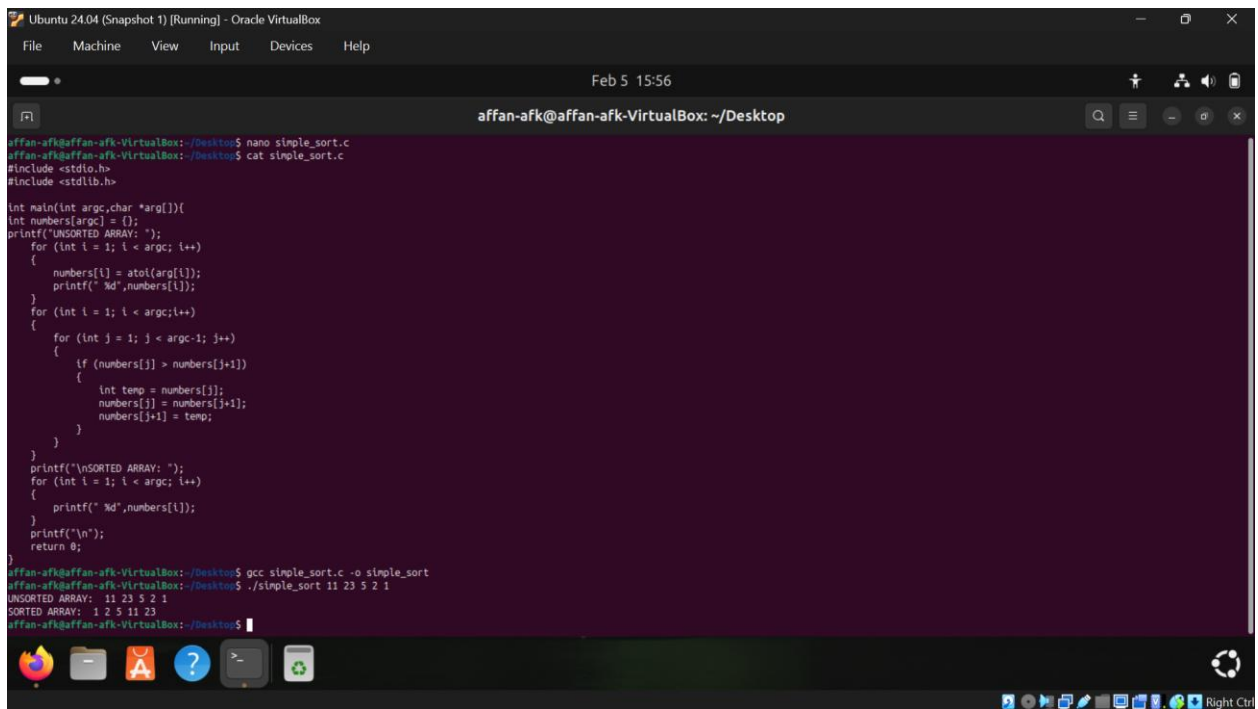


TASK 1:



```
affan-afk@affan-afk-VirtualBox: ~/Desktop$ nano grade_calculator.c
affan-afk@affan-afk-VirtualBox: ~/Desktop$ cat grade_calculator.c
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *arg[]){
    int grades = 0;
    int count = argc-1;
    for(int i = 1; i<argc; i++){
        if(atoi(arg[i]) >= 0 && atoi(arg[i]) <= 100){
            grades += atoi(arg[i]);
        }
        else{
            printf("Value %d is invalid, and disregarded\n", i);
            count--;
        }
    }
    float average = (float)grades/(count);
    printf("Average of %d students is : %f\n", count, average);
    return 0;
}
affan-afk@affan-afk-VirtualBox: ~/Desktop$ gcc grade_calculator.c -o grade_calculator
affan-afk@affan-afk-VirtualBox: ~/Desktop$ ./grade_calculator 85 90 78
Average of 3 students is : 84.333336
affan-afk@affan-afk-VirtualBox: ~/Desktop$ ./grade_calculator -2 56 607
Value 1 is invalid, and disregarded
Value 3 is invalid, and disregarded
Average of 1 students is : 56.000000
affan-afk@affan-afk-VirtualBox: ~/Desktop$ ./grade_calculator 88 56 67
Average of 3 students is : 70.333336
affan-afk@affan-afk-VirtualBox: ~/Desktop$
```

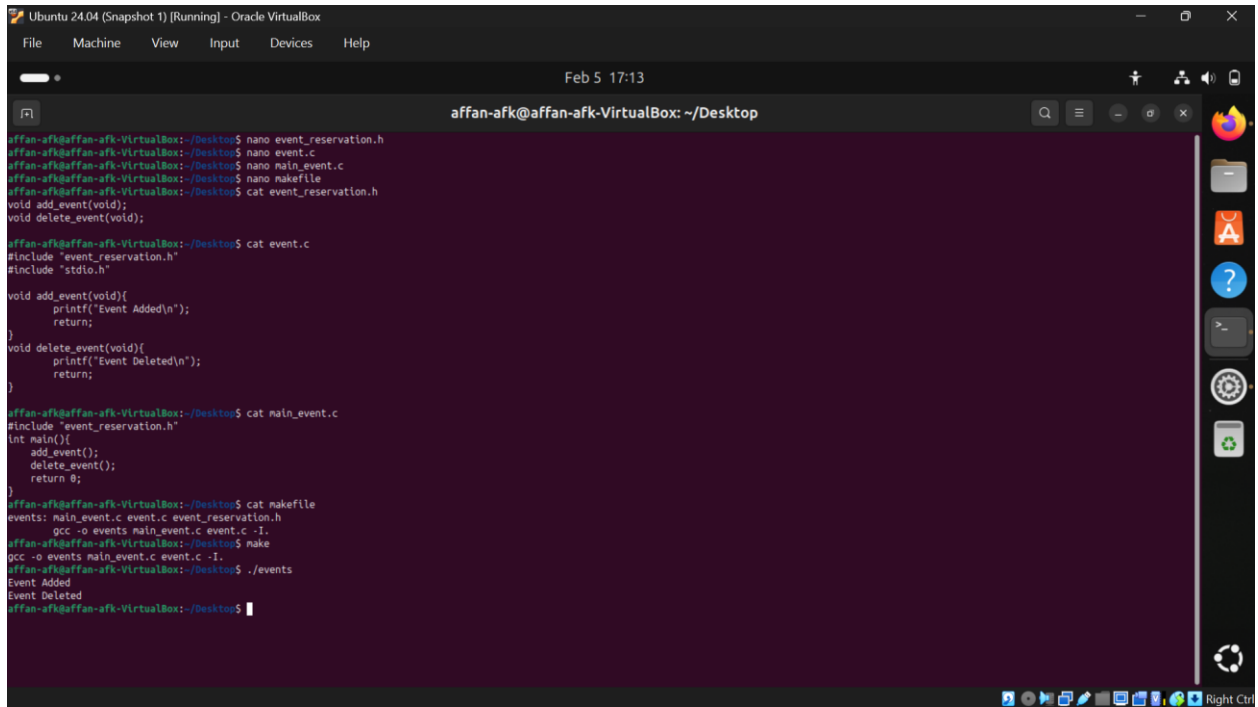
TASK 2:



```
affan-afk@affan-afk-VirtualBox: ~/Desktop$ nano simple_sort.c
affan-afk@affan-afk-VirtualBox: ~/Desktop$ cat simple_sort.c
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *arg[]){
    int numbers[argc] = {};
    printf("UNSORTED ARRAY: ");
    for (int i = 1; i < argc; i++)
    {
        numbers[i] = atoi(arg[i]);
        printf(" %d", numbers[i]);
    }
    for (int i = 1; i < argc; i++)
    {
        for (int j = i; j < argc-1; j++)
        {
            if (numbers[j] > numbers[j+1])
            {
                int temp = numbers[j];
                numbers[j] = numbers[j+1];
                numbers[j+1] = temp;
            }
        }
    }
    printf("\nSORTED ARRAY: ");
    for (int i = 1; i < argc; i++)
    {
        printf(" %d", numbers[i]);
    }
    printf("\n");
    return 0;
}
affan-afk@affan-afk-VirtualBox: ~/Desktop$ gcc simple_sort.c -o simple_sort
affan-afk@affan-afk-VirtualBox: ~/Desktop$ ./simple_sort 11 23 5 2 1
UNSORTED ARRAY: 11 23 5 2 1
SORTED ARRAY: 1 2 5 11 23
affan-afk@affan-afk-VirtualBox: ~/Desktop$
```

TASK 3:



```
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano event_reservation.h
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano event.c
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano main_event.c
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano makefile
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat event_reservation.h
void add_event(void);
void delete_event(void);

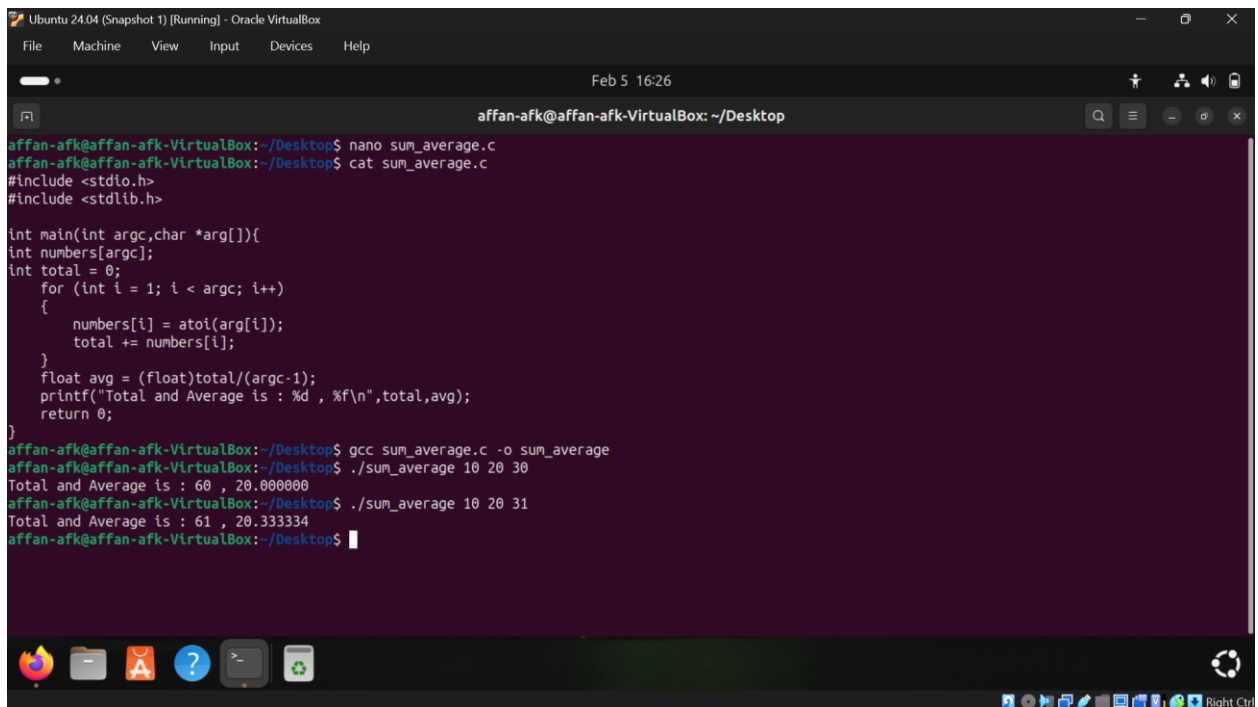
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat event.c
#include "event_reservation.h"
#include "stdio.h"

void add_event(void){
    printf("Event Added\n");
    return;
}
void delete_event(void){
    printf("Event Deleted\n");
    return;
}

affan-afk@affan-afk-VirtualBox:~/Desktop$ cat main_event.c
#include "event_reservation.h"
int main(){
    add_event();
    delete_event();
    return 0;
}

affan-afk@affan-afk-VirtualBox:~/Desktop$ cat makefile
events: main_event.c event.c event_reservation.h
    gcc -o events main_event.c event.c -I.
affan-afk@affan-afk-VirtualBox:~/Desktop$ make
gcc -o events main_event.c event.c -I.
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./events
Event Added
Event Deleted
affan-afk@affan-afk-VirtualBox:~/Desktop$
```

TASK 4:

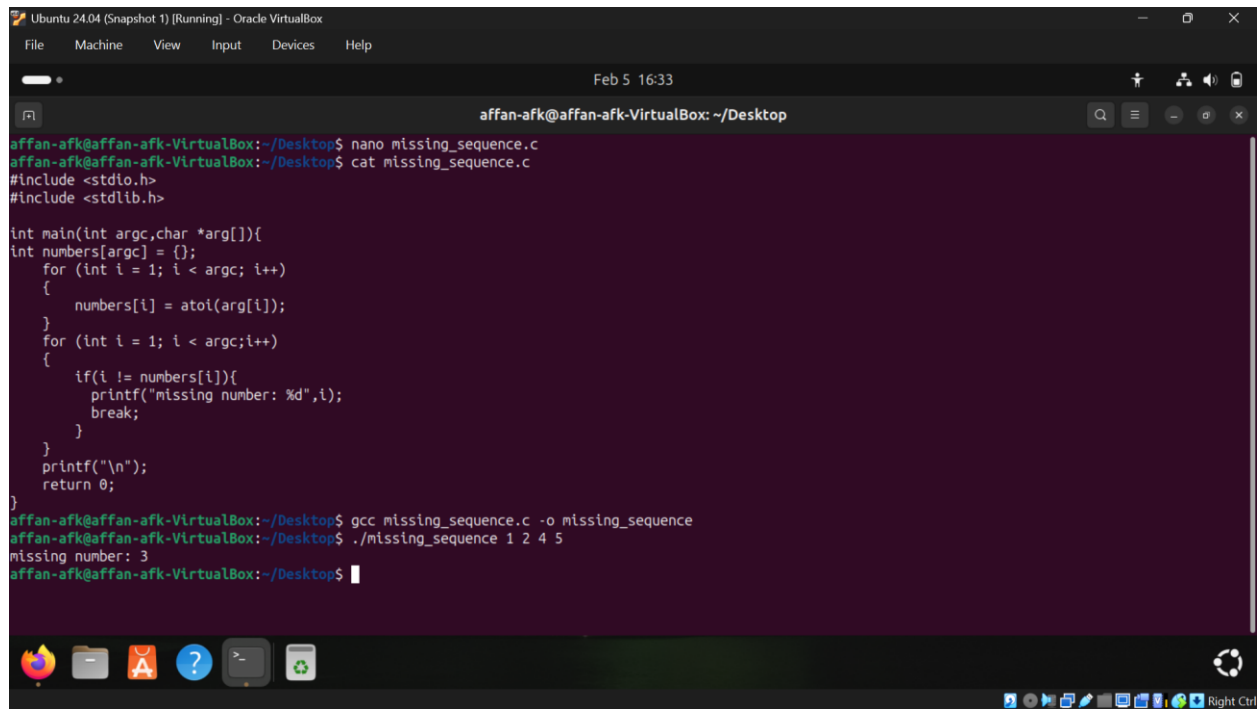


```
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano sum_average.c
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat sum_average.c
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *arg[]){
    int numbers[argc];
    int total = 0;
    for (int i = 1; i < argc; i++)
    {
        numbers[i] = atoi(arg[i]);
        total += numbers[i];
    }
    float avg = (float)total/(argc-1);
    printf("Total and Average is : %d , %f\n", total, avg);
    return 0;
}

affan-afk@affan-afk-VirtualBox:~/Desktop$ gcc sum_average.c -o sum_average
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./sum_average 10 20 30
Total and Average is : 60 , 20.000000
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./sum_average 10 20 31
Total and Average is : 61 , 20.333334
affan-afk@affan-afk-VirtualBox:~/Desktop$
```

TASK 5:



```
affan-afk@affan-afk-VirtualBox: ~/Desktop
affan-afk@affan-afk-VirtualBox:~/Desktop$ nano missing_sequence.c
affan-afk@affan-afk-VirtualBox:~/Desktop$ cat missing_sequence.c
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *arg[]){
int numbers[argc] = {};
for (int i = 1; i < argc; i++)
{
    numbers[i] = atoi(arg[i]);
}
for (int i = 1; i < argc; i++)
{
    if(i != numbers[i]){
        printf("missing number: %d", i);
        break;
    }
}
printf("\n");
return 0;
}
affan-afk@affan-afk-VirtualBox:~/Desktop$ gcc missing_sequence.c -o missing_sequence
affan-afk@affan-afk-VirtualBox:~/Desktop$ ./missing_sequence 1 2 4 5
missing number: 3
affan-afk@affan-afk-VirtualBox:~/Desktop$
```